Fisheries and Oceans Canada
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Ecosystems and Oceans Science
Sciences des écosystèmes et des océans

GOTeDNA

OBIS
OCEAN BIODIVERSITY INFORMATION SYSTEM

Standard Operating Procedure (SOP) for data compilation and reporting eDNA metabarcoding data to OBIS format for GOTeDNA App extraction and functionality

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Purpose: To merge formatting guidelines for eDNA data (metabarcoding) from the OBIS manual with requirements for the GOTeDNA App for optimal functionality to inform best practices for data collection. This reporting uses all mandatory terms provided by the Data Formatting Guidelines to Make eDNA FAIR (Findable, Accessible, Interoperable, and Reusable) using the most up-to-date FAIR eDNA (FAIRe) Metadata Checklist v1.0.2 (Takahashi et al., 2025). Major requirements for GOTeDNA includes data with present and absent values in the Occurrence Core file, in addition to the OBIS extensions "DNA derived Data" and "extended Measurement or Fact".

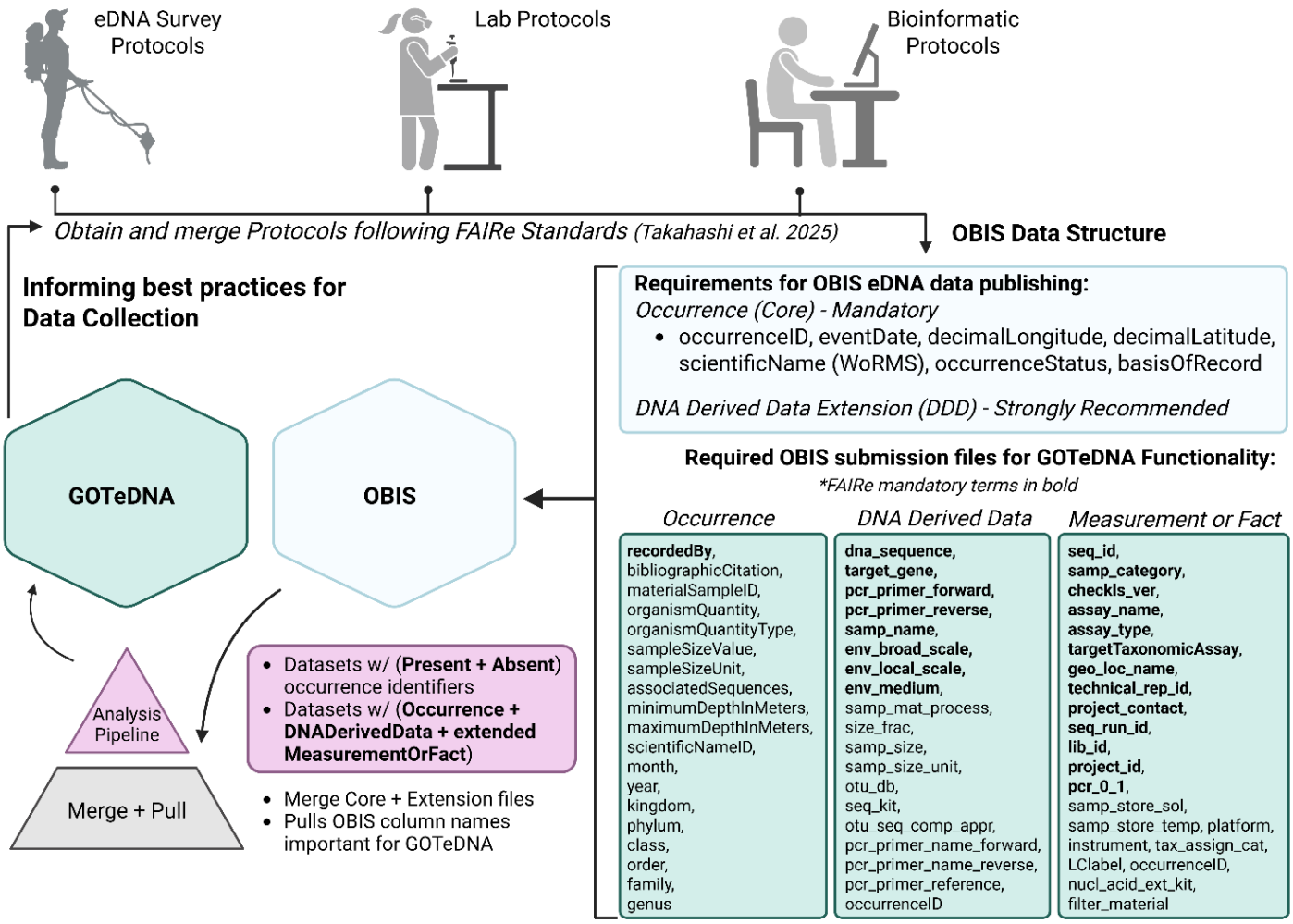


Figure 1. Visual workflow presenting a broad summary of how data is compiled for the GOTeDNA app. First, data and protocols from field surveys, lab, and bioinformatics are obtained, merged together and standardized to term names from the FAIRe Checklist. Minimum requirements for publishing genetic data within OBIS include a core Occurrence table and a DNA derived data (DDD) extension. Within the Occurrence core file,

occurrenceStatus is mandatory to include present/absent values to calculate detection probability within the GOTeDNA App. To be consistent with FAIRe standards, BeBOP protocols, and GOTeDNA nomenclature, the extended Measurement or Fact (eMOF) extension is required to utilize term names outside of Darwin Core (DwC) terms. Each core and extension file are inputted into their affiliated Country's OBIS Integrated Publishing Toolkit (IPT) to be publicly available. From this publicly available data, GOTeDNA filters datasets from OBIS that have each Occurrence core, DDD, and eMOF extension, as well as both present and absent occurrence values. These three files are then merged so that GOTeDNA can pull the column names valuable to run statistics within the app and ultimately inform best practices for eDNA data collection for a circular data collection, management, and information process.

Resources:
OBIS Manual - The OBIS Manual
Darwin Core Quick Reference Guide (OBIS Definitions) - Darwin Core Quick Reference Guide - Darwin Core
FAIRe Checklist (FAIRe Definitions) - Download - FAIR eDNA

Summary of GOTeDNA Minimum Requirements:

1. Both present (>0 reads) and absent (0 reads) metabarcoding data

2. Three excel sheets:
    a. Occurrence
    b. DNA Derived Data
    c. extension Measurement or Fact (formatted differently, see 13.4 How to format extendedMeasurementOrFact tables | The OBIS Manual)

3. For functionality, GOTeDNA specifically requires "samp_name", "target_gene", "organismQuantity", "pcr_primer_name_forward", "pcr_primer_name_reverse"

Datasets in OBIS that do not meet these requirements will be filtered out by GOTeDNA and cannot be accessed by the app.

Procedure:

1. **Format your data in excel templates with the terms provided (but not limited to) for each of the three "Occurrence", "DNA Derived Data extension", and "extended Measurement Or Fact extension" files.**

A glossary is provided at the end of this document for term names for each file. When a clear definition relating to Genetic data is not available in OBIS, we provide the definition from FAIRe or GOTeDNA

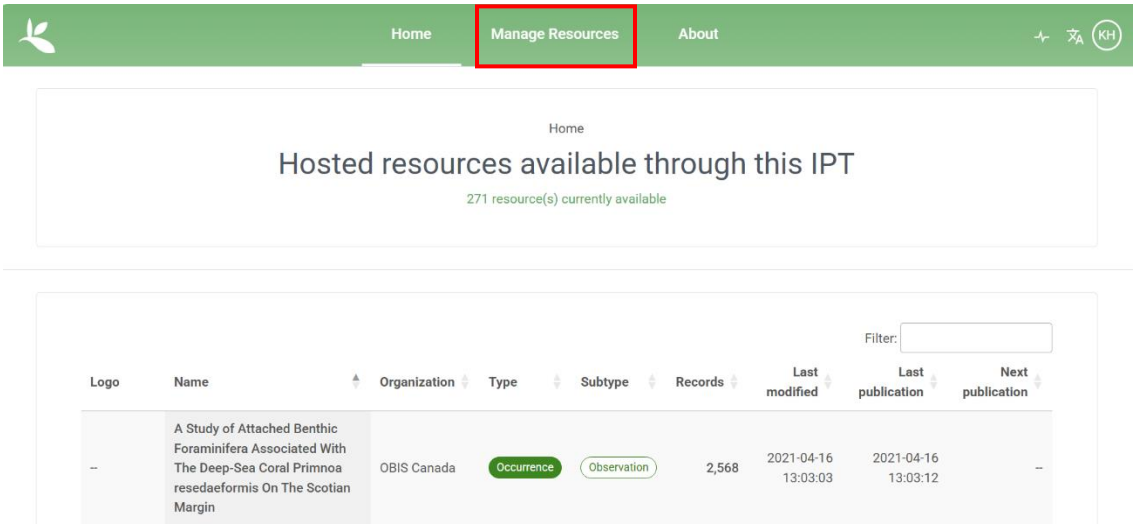
Example R code to format data into the three excel sheets for OBIS submission have been developed for use with the "GOTeDNA Sample Template" associated with this document: *gotedna/OBIS at main · GOTeDNA-OBON/gotedna*

NOTE: The code provides space to add and merge two additional excel files that will likely be obtained from 1) the group that conducted sequencing (for term names "seq_run_id" and "lib_id", 2) the group that conducted bioinformatic analysis (for term names "seq_id" and dna_sequence"). Edit the code as you see fit to accommodate those files.

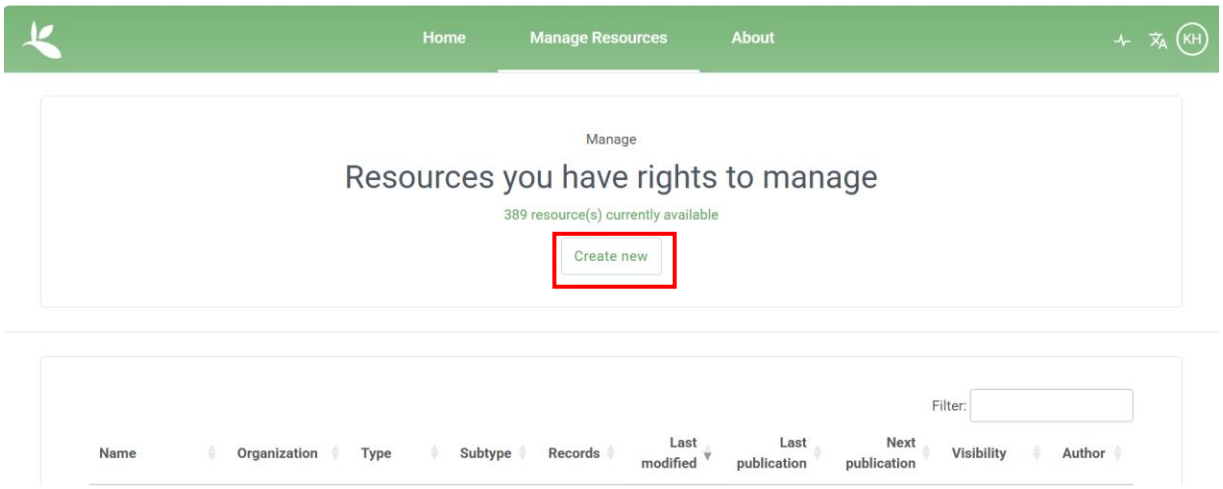
2. **Identify and contact your OBIS node manager**

You can identify your OBIS node Manager here: Contact | Ocean Biodiversity Information System (OBIS). Contact your Node Manager who will create an IPT account for you and provide details to log in and change your password.

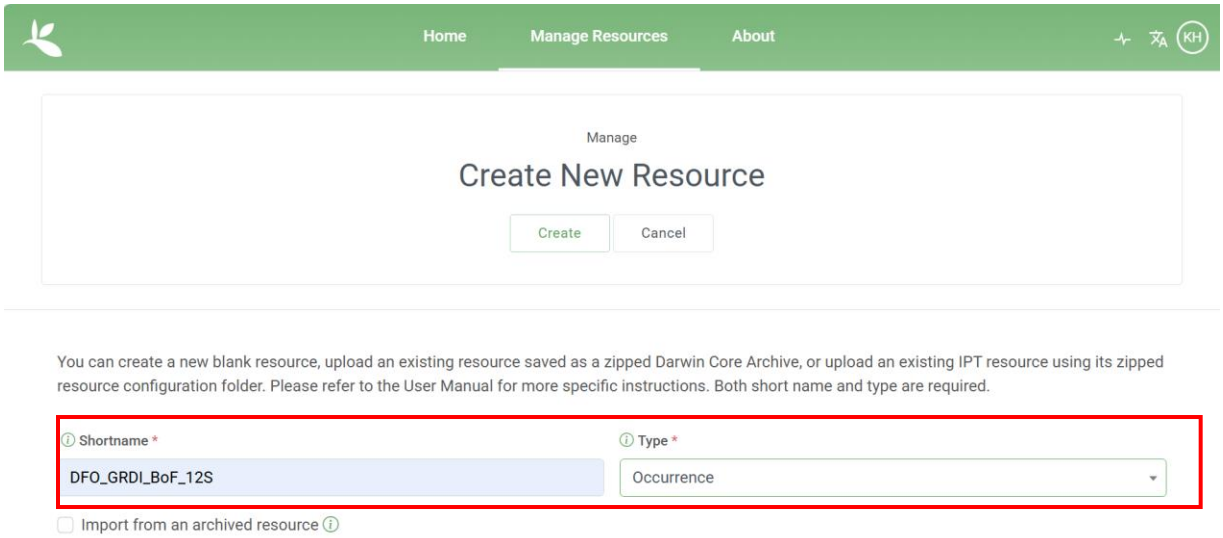
3. **Go to "Manage Resources"**



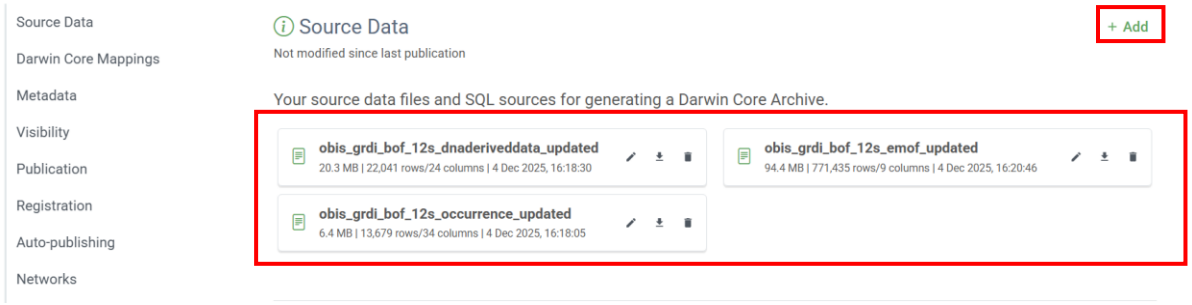
4. **Create a new Resource**



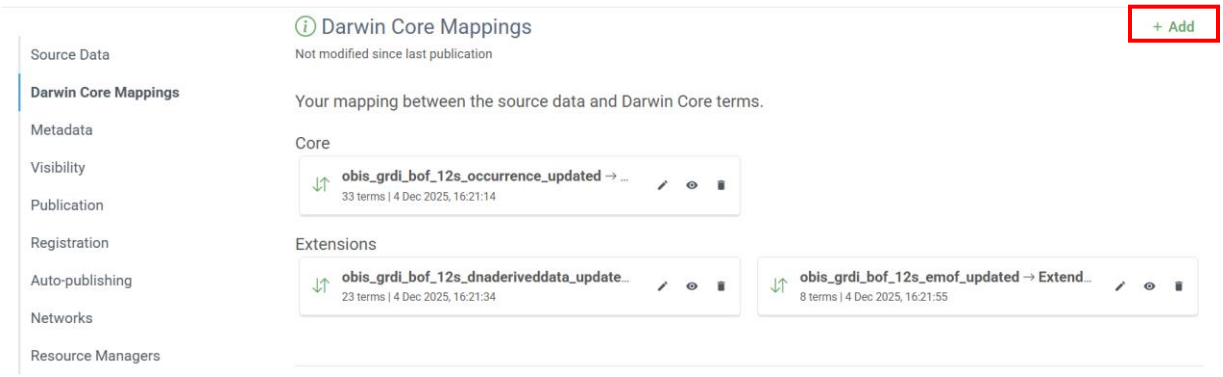
5. **Under "Shortname", provide a short name describing your dataset and select "Occurrence" under the "Type" dropdown**



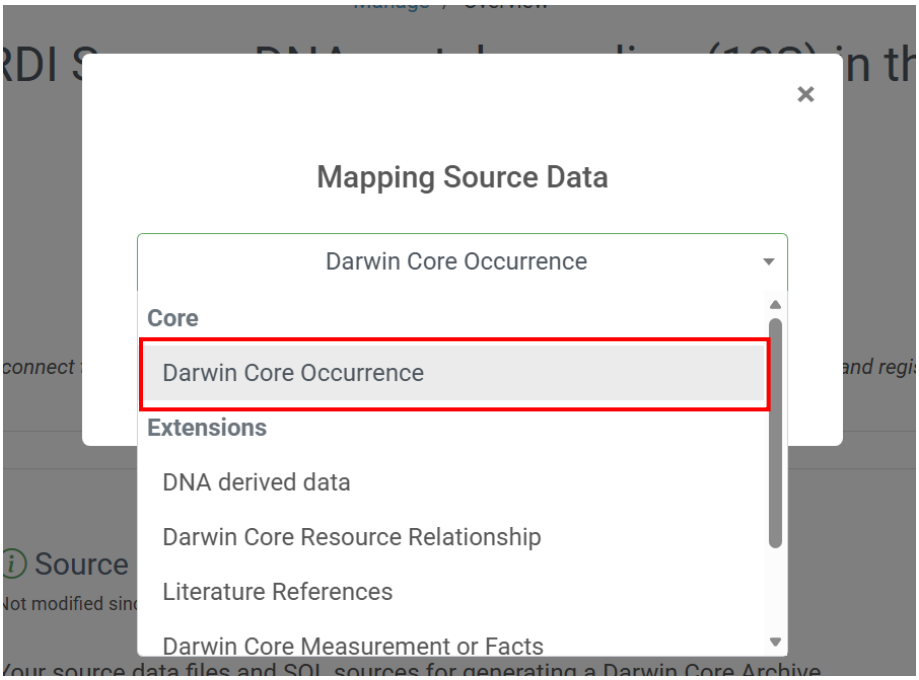
6. **Once a new resource is created, you will upload your prepared .csv files to "Source Data"**



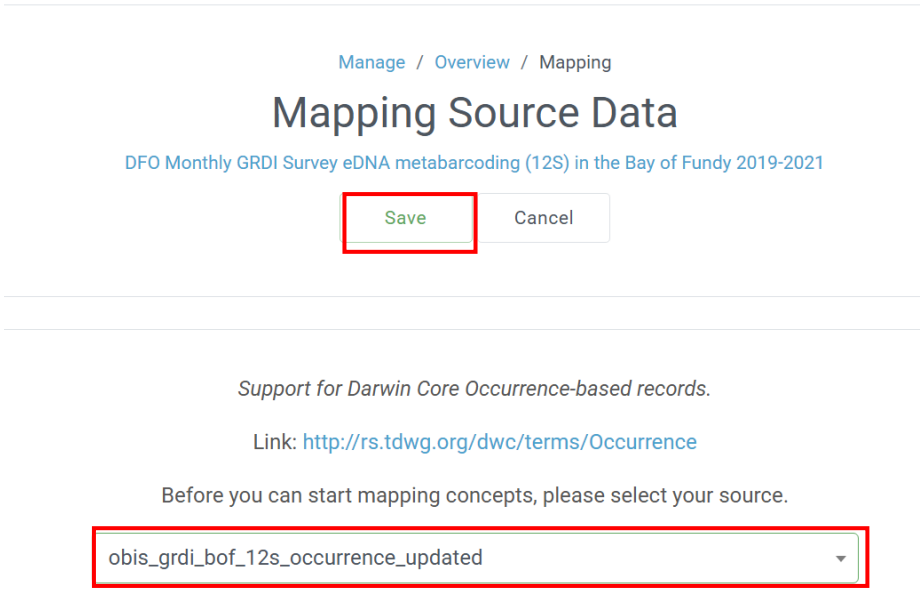
7. **Map the Source Data to Darwin Core terms**
    a. **Add the type of source data you want to map (core occurrence, extensions)**



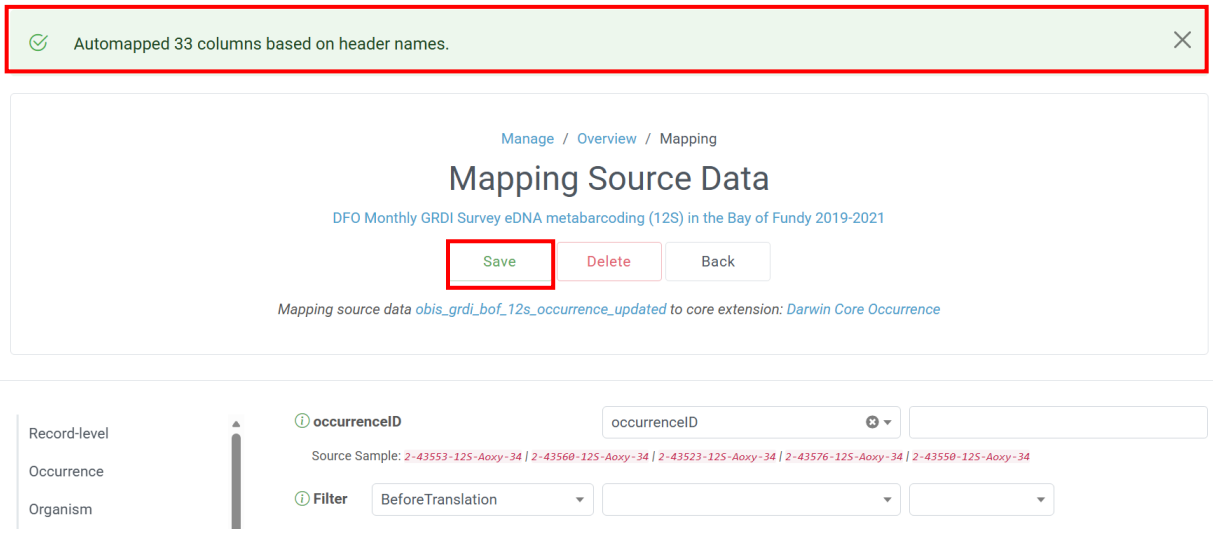
    b. **Start by selecting the Occurrence Source Data**



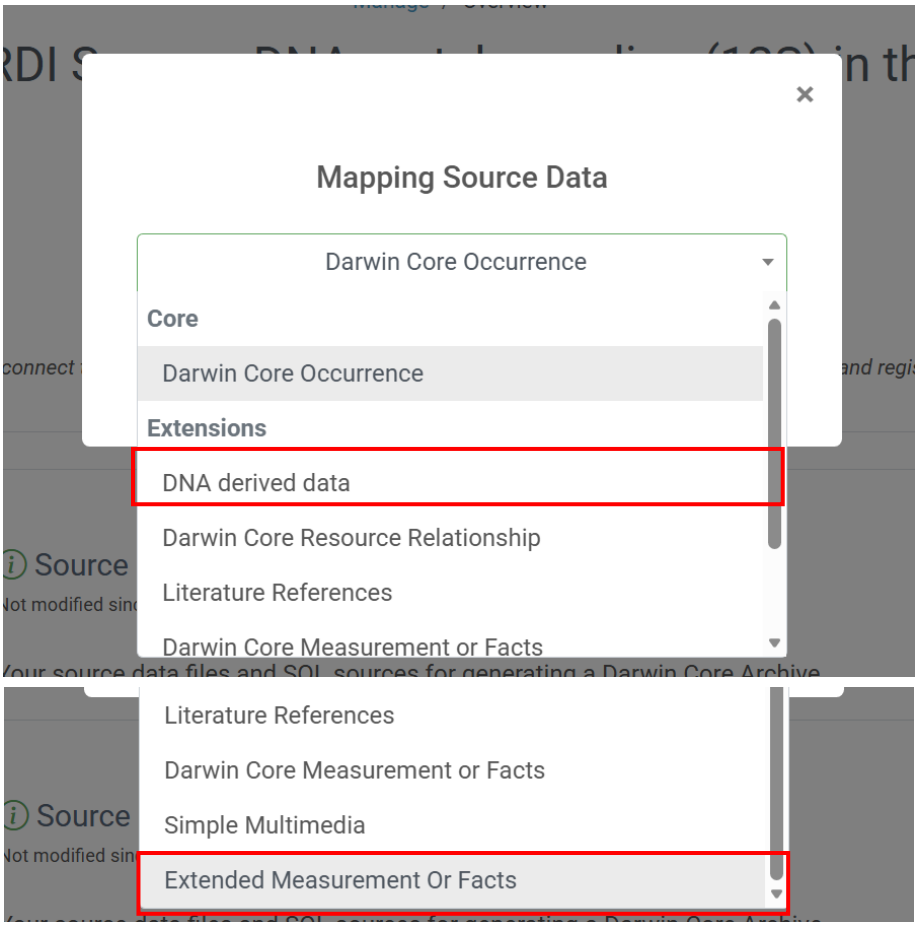
c. Then select your Occurrence csv file



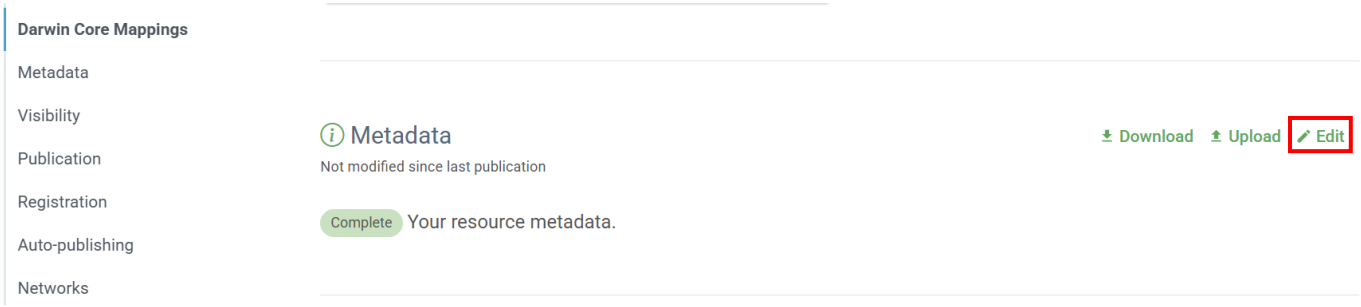
d. The column headings should then be automapped based on header names, given the column headings are the same as those used in this document



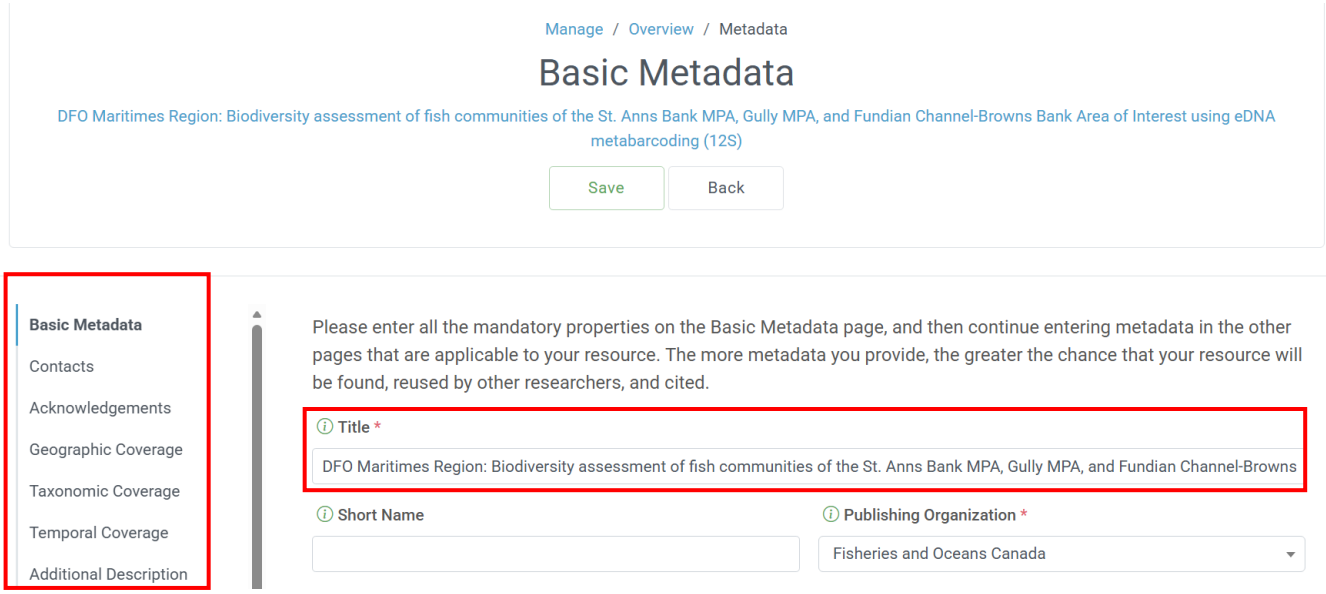
Note: After mapping the Darwin Core Occurrence, repeat these steps for the extension files



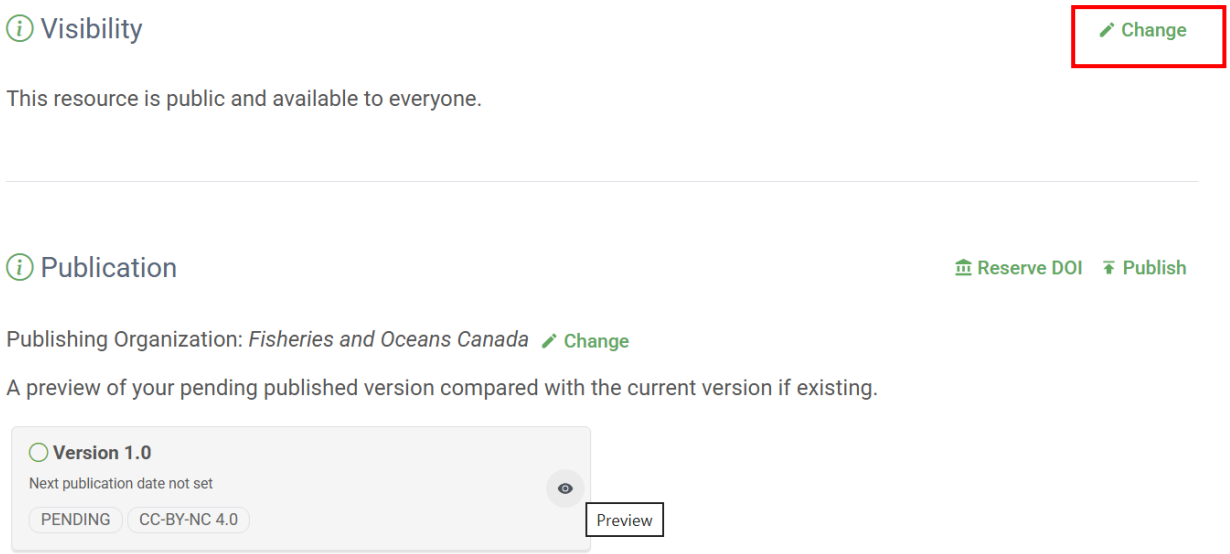
8. **Return to the main resource page (select the "back button" if on the mapping page) to fill in Mandatory Metadata information**



a. **Input information for each section that has a red asterisk (*). Then, navigate through the tabs on the left hand side to continue filling out sections for the metadata**

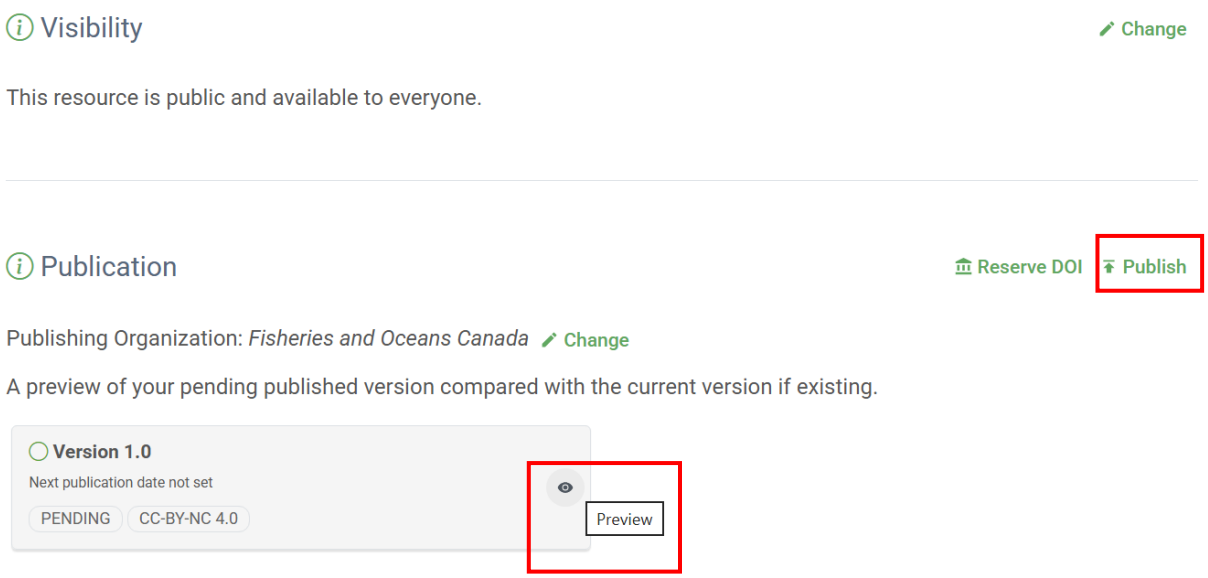


9. **Return to the main resource page and adjust the visibility of the dataset to Public (will be on private originally) and preview the dataset before publishing**

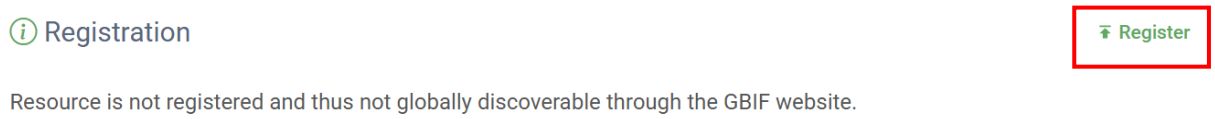


10. **After previewing the dataset, you can return to any of the sections previously listed and make adjustments as necessary. Once happy with the dataset, you can publish the data**

Note: You can always go back and edit your dataset later with additional versions



11. **After publishing, you can register the dataset to GBIF on the main resource page so that it is globally discoverable across both GBIF and OBIS open-access data platforms**



Glossary

Occurrence (Core)

OBIS currently has seven required and one strongly recommended DwC term:

Definitions obtained from OBIS:

occurrenceID - (required term) is an identifier for the occurrence record and should be persistent and globally unique. If the dataset does not yet contain (globally unique) occurrenceIDs, then they should be created. Guideline for ID creation can be found here

occurrenceStatus - (required term) is a statement about the presence or absence of a taxon at a location. It is an important term, because it allows us to distinguish between presence and absence records. It is a required term and should be filled in with either present or absent.

basisOfRecord - (required term) specifies the nature of the record, i.e. whether the occurrence record is based on a stored specimen or an observation. For records pertaining to genetic samples, basisOfRecord can be MaterialSample (e.g. in the DNA-derived data extension).

scientificName - (required) should always contain the originally recorded full scientific name, even if the name is currently a synonym. This is necessary to be able to track back records to the original dataset. The name should include authorship and date information if known and should be the lowest possible taxonomic rank that can be determined, preferably at species level or lower, but higher ranks, such as genus, family, order, class etc. are also acceptable.

eventDate - (required) The date and time at which an occurrence was recorded. This term uses the ISO 8601 standard and OBIS recommends using the extended ISO 8601 format with hyphens. Example: 2013-02-27

More specific guidelines on formatting dates and times can be found in the Common Data formatting issues page

decimalLatitude - (required) values range from -90 to 90, with positive values indicating locations north of the Equator and negative values indicating locations south of it.

decimalLongitude - (required) values range from -180 to 180, with positive values representing locations east of the Greenwich Meridian and negative values representing locations west of it.

scientificNameID – (Strongly recommended) A WoRMS LSID. The LSID used for scientificNameID should correspond to the name provided to the scientificName field, even if it is not the currently accepted scientific name. OBIS will use this identifier to pull the taxonomic information from the World Register of Marine Species (WoRMS) into OBIS and attach it to your dataset. This information includes:

- Taxonomic classification (kingdom through species)
- The accepted name in case of invalid names or synonyms
- AphiaID
- IUCN red list category

LSIDs are persistent, location-independent, resource identifiers for uniquely naming biologically significant resources. More information on LSIDs can be found at www.lsid.info.

bibliographicCitation – A bibliographic reference for the resource.

associatedSequences - global unique identifiers, URLs, or URIs pointing to respectively associated media (e.g. online image or video), associated literature (e.g. DOIs) or genetic sequence information (e.g. GenBANK ID). It is recommended to ensure URLs include the domain name (e.g. NCBI) when possible.

eventID - an identifier for the sampling or observation event.

recordedBy – (Mandatory for FAIRe) can hold a list (concatenated and separated) of names of people, groups, or organizations responsible for recording the original Occurrence. The primary collector or observer, especially one who applies a personal identifier (recordNumber), should be listed first.

minimumDepthInMeters - this should be the depth at which the **sample was taken** and not the water column depth at that location. The lesser depth of a range of depth below the local surface, in meters.

maximumDepthInMeters - this should be the depth at which the **sample was taken** and not the water column depth at that location. The greater depth of a range of depth below the local surface, in meters.

month – the integer month in which the dwc:event occurred

year – the four digit year in which the dwc:event occurred, according to the Common Era Calendar

Definitions obtained by FAIRe:

materialSampleID – An identifier for the physical sample (as opposed to the extracted DNA sample defined by samp_name). In the absence of a persistent global unique identifier, construct one from a combination of identifiers in the record.

Definitions obtained by GOTeDNA:

organismQuantity – Number of reads (OTU or ASV) in the sample.

organismQuantityType - Should always be "DNA sequence reads".

sampleSizeValue - Total number of reads in the sample. This is important since it allows calculating the relative abundance of each OTU or ASV within the sample. This number should preferably be calculated after universal processing (quality control, ASV denoising, chimera removal, etc.), but before manual/selective removal of e.g. non-target OTUs or ASVs from the dataset. Rarefaction (resampling to even sequencing depth across samples) is not necessary or advised.

sampleSizeUnit - Should always be "DNA sequence reads".

DNA Derived Data (Extension)

Definitions obtained from OBIS:

dna_sequence – (Mandatory for FAIRe) is the most important field, where the ASV/OTU sequence will be recorded. This field can then be searched with sequence alignment methods to, for example, find closely related sequences recorded in other studies, and will allow very powerful data comparison and analysis in the future. It will also make your sequence available in the OBIS sequence search tool.

occurrenceID - is an identifier for the occurrence record and should be persistent and globally unique. If the dataset does not yet contain (globally unique) occurrenceIDs, then they should be created. Guideline for ID creation can be found here

Definitions obtained from FAIRe:

target_gene - Targeted gene or locus name for marker gene studies

Controlled Vocabulary: 12S rRNA (SSU mitochondria) | 16S rRNA (LSU mitochondria) | 16S rRNA (SSU prokaryote) | 23S rRNA (LSU prokaryote) | 18S rRNA (SSU eukaryote) | 28S rRNA

(LSU eukaryote) | rbcL | CytB | COI | COII | COIII | nifH | ITS | ND1 | ND2 | ND3 | ND4 | ND5 | ND6 | amoA | rpoB | rpoC1 | rpoC2 | matK | trnH | trnL | psbK | D-loop | other:

pcr_primer_forward –(Mandatory for FAIRe) Forward PCR primer (5' - 3') that were used to amplify the sequence of the targeted gene. The primer sequence should NOT contain MIDs and adapter sequences, and should be reported in uppercase letters. If multiple forward primers are present in a single PCR reaction, either report them separately in separate submissions (with separate OTU tables for each primer set), or list the primers here separated by pipe (|) and provide one OTU table resulting from all the primers applied here. The latter is recommended only when the multiple primers target the same gene region and thus a single OTU table was generated from the multiplexed PCR. This term is required except for PCR free methods.

pcr_primer_reverse –(Mandatory for FAIRe) Reverse PCR primer (5' - 3') that were used to amplify the sequence of the targeted gene. The primer sequence should NOT contain MIDs and adapter sequences, and should be reported in uppercase letters. If multiple reverse primers are present in a single PCR reaction, either report them separately in separate submissions (with separate OTU tables for each primer set), or list the primers here separated by pipe (|) and provide one OTU table resulting from all the primers applied here. The latter is recommended only when the multiple primers target the same gene region and thus a single OTU table was generated from the multiplexed PCR. This term is required except for PCR free methods.

pcr_primer_name_forward - Name of the pcr_primer_forward. If multiple primers are listed in pcr_primer_forward, list the names respectively, separated by pipe (|).

pcr_primer_name_reverse - Name of the pcr_primer_reverse. If multiple primers are listed in pcr_primer_reverse, list the names respectively, separated by pipe (|).

pcr_primer_reference - Reference (i.e. DOI) of the pcr_primer_forward. If multiple primers are listed in pcr_primer_forward, list the references respectively, separated by pipe (|).

samp_name – (Mandatory by FAIRe) Sample Name is a name that you choose for the sample. It can have any format, but we suggest that you make it concise, unique and consistent within your lab, and as informative as possible. Every Sample Name from a single Submitter must be unique.

samp_collec_device - The device used to collect an environmental sample. This term accepts terms listed under environmental sampling device (https://ontobee.org/ontology/OBI?iri=http://purl.obolibrary.org/obo/OBI_0000968) such as syringe [OBI_0000422] and container [OBI_0000967] as well as free-text entry (i.e. niskin bottle, push core, spoon, water bottle).

size_frac - Filtering pore size used in sample filtration. Unit = µm

samp_size – Amount or size of original sample (volume, mass or area) that was collected from the environment. Specify the unit under the term samp_size_unit. Note: the amount or size of (sub)sample processed for DNA extraction should be entered under the term samp_vol_we_dna_ext.

samp_size_unit - Unit for samp_size

Controlled Vocabulary: mL | L | mg | g | kg | cm2 | m2 | cm3 | m3 | other:

samp_mat_process - Any processing applied to the sample during or after retrieving the sample from environment (i.e. sieving, filtration, precipitation, homogenising, subsampling).

seq_kit – The name of sequencing kit

project_name - Name of the project within which the sequencing was organized

env_broad_scale – (Mandatory for FAIRe) Major environmental system your sample came from. The systems identified should have a coarse spatial grain, to provide the general environmental context of where the sampling was done. Select from ENVO's biome class: http://purl.obolibrary.org/obo/ENVO_00000428. Examples include; estuarine biome [ENVO:01000020], tropical moist broadleaf forest biome [ENVO:01000228]

env_local_scale - (Mandatory for FAIRe) The entity or entities which are in your sample or specimen's local vicinity and which you believe have significant causal influences on your sample. Select from ENVO's layer class: http://purl.obolibrary.org/obo/ENVO_01000281. Examples include; marine pelagic zone [ENVO:00000208], estuarine coastal surface layer [ENVO:01001302]

env_medium - (Mandatory for FAIRe) Environmental material or materials (pipe separated) immediately surrounded your sample prior to sampling. Select from ENVO's environmental material class (entity/ continuant/ independent continuant/ material entity/ environmental material): http://purl.obolibrary.org/obo/ENVO_00010483. Examples include; soil [ENVO:00001998], marine sediment [ENVO:03000033], air [ENVO:00002005], sea water [ENVO:00002149], fresh water [ENVO:00002011], tap water [ENVO:00003096], fecal material [ENVO:00002003], digestive tract environment [ENVO:2100002], intestine environment [ENVO:2100002], planktonic material [ENVO:01000063]

otu_seq_comp_appr - Alignment tool/software used for taxonomic assignment

otu_db - List of reference database(s) (i.e. sequences not generated as part of the current study) along with version number, that were used to assign taxonomy to OTUs or ASVs. If a custom database was used, enter 'custom' here, and provide the information of the database under the term otu_db_custom

Extended Measurement or Fact (Extension)

Note: Format for the extended Measurement or Fact template is different than the core and DNA derived templates

Please find details on how to format extendedMeasurementOrFact tables here: 13.4 How to format extendedMeasurementOrFact tables | The OBIS Manual

Definitions obtained from OBIS:

occurrenceID - (required term) is an identifier for the occurrence record and should be persistent and globally unique. If the dataset does not yet contain (globally unique) occurrenceIDs, then they should be created. Guideline for ID creation can be found here

measurementType - The nature of the measurement, fact, characteristic, or assertion.
Note: this is where you can use FAIRe terms not in DarwinCore

Recommended best practice is to use a controlled vocabulary. This term has an equivalent in the dwciri: namespace that allows only an IRI as a value, whereas this term allows for any string literal value

measurementTypeID – An identifier for the measurementType (global unique identifier, URI). The identifier should reference the measurementType in a vocabulary
NOTE: this is where you can put the URL links;
FAIRe: https://github.com/FAIR-eDNA/FAIRe_checklist
GOTeDNA: https://github.com/GOTeDNA-OBON

measurementValue - The value of the measurement, fact, characteristic, or assertion.
Note: this is where the value from the FAIRe terms will go

Definitions obtained from FAIRe:

seq_id – (Mandatory) Query sequence ID. Must be unique within the dataset

samp_store_sol – (Highly recommended) Solution within which the sample was stored. Note: If a sample was processed before DNA extraction (e.g., filtration, homogenisation, subsampling), enter the storage solution before the process here, and the storage solution of the processed sample under the term

Controlled Vocabulary: ethanol | sodium acetate | longmire | lysis buffer | none | other:

samp_store_temp – (Highly recommended) Temperature at which the sample was stored. Enter either the fixed term "ambient temperature" or a numeric value. Note: If a sample was processed before DNA extraction (e.g., filtration, homogenisation, subsampling), enter the storage temperature before the process here, and the storage temperature of the processed sample (i.e., filter paper) under a term prepped_samp_store_temp. Unit = degree Celsius

samp_category – (Mandatory) A negative control type used in an investigation. See Table 2 in Klymus et al. 2024 (DOI: 10.3897/mbmg.8.128689) for the description of each control type, how and why it's created. For example, process negative is a sample added during processing that lacks target DNA, such as blank filter papers.

Controlled Vocabulary: sample | negative control | positive control | PCR standard | other:

> Note: If samp_category = negative control or If samp_category = positive control, the terms **neg_cont_type** and **pos_cont_type** are Mandatory

checkls_ver – (Mandatory) The version number of the FAIR eDNA (FAIRe) metadata checklist used to format the data. Example: v1.0.2

assay_name – (Mandatory) A brief, concise identifier for assay with no spaces or special characters, ensuring machine readability. This ID will be used in file names as 'assay_name'.

assay_type – (Mandatory) The method used in the eDNA study to detect taxon/taxa of interest in environmental sample. A targeted eDNA assay is designed to detect the presence of a specific species or a small subset of taxa (such as a genus or family) using techniques like quantitative PCR (qPCR). This contrasts with a metabarcoding eDNA assay which aims to characterize the composition of broader taxonomic communities, often encompassing multiple phyletic groups, by utilising high-throughput sequencing such as metabarcoding.

Controlled Vocabulary: targeted | metabarcoding | other:

targetTaxonomicAssay – (Mandatory) Taxa or species name targeted by the primers, probes, and/or other approaches applied in the PCR. Example: verteberate

geo_loc_name - (Mandatory) The geographical origin of the sample as defined by the country or sea name followed by regions and localities. Country or sea names should be chosen from the INSDC Geographic Location Name List INSDC country list (http://insdc.org/country.html). Fixed format following <country or sea names>:<region>,<locality>. E.g., "USA: Maryland, Bethesda" , "Atlantic Ocean:Charlie Gibbs Fracture Zone"

technical_rep_id – (Mandatory) An integer uniquely identifying a specific PCR technical replicate within a sample.

project_contact – (Mandatory) A contact person and their contact (i.e., email address) for data enquiry

seq_run_id – (Mandatory) A brief, concise identifier for sequencing run with no spaces or special characters, ensuring machine readability. This ID will be used in file names as 'seq_run_id'.

lib_id – (Mandatory) An identifier of a library (an amplicon of a sample with unique MIDs)

project_id – (Mandatory) A brief, concise project identifier with no spaces or special characters, ensuring machine readability. This ID will be used in data file names as 'project_id'.

pcr_0_1 – (Mandatory) Was PCR performed in the study? Yes = 1, No (pcr-free approaches including shotgun metagenomics, Nanopore) = 0

nucl_acid_ext – (Highly recommended) A link to a literature reference, electronic resource or a standard operating procedure (SOP), that describes the material separation to recover the nucleic acid fraction from a sample. If a custom extraction method has been developed and applied to the samples within this dataset, it is Highly recommended to make the SOP available and provide the link here.

nucl_acid_ext_kit – (Highly recommended) The name of the extraction kit used to recover the nucleic acid fraction of an input material. Example: Qiagen PowerSoil Kit

platform – (Highly recommended) General sequencing platform used.

Controlled Vocabulary: ILLUMINA | BGISEQ | CAPILLARY | DNBSEQ | ELEMENT | GENAPSYS | GENEMIND | HELICOS | ION_TORRENT | LS454 | OXFORD_NANOPORE | PACBIO_SMRT | TAPESTRI | VELA_DIAGNOSTICS | ULTIMA | other:

instrument – (Highly recommended) Sequencing instrument (manufacturer and model) used. Where possible the term should be taken from the OBI list of DNA sequencers (https://ontobee.org/ontology/OBI?iri=http://purl.obolibrary.org/obo/OBI_0002750)

tax_assign_cat – (Highly recommended) High-level category of taxonomic assignment approach. Examples using sequence similarity: BLAST/MegaBLAST, USEARCH/VSEARCH. Examples using sequence composition: naive Bayes classifier, CLARK, Kraken. Examples using phylogeny: SEPP, EPA-ng, pplacer. Examples using probabilistic: PROTAX, TIPP.

Controlled Vocabulary: sequence similarity | sequence composition | phylogeny | probabilistic | other:

filter_material – (Highly recommended) Material of filter

Controlled Vocabulary: cellulose | cellulose ester | glass fiber | thermoplastic membrane | track etched polycarbonate | nylon | polyethersulfone| other:

<u>Definitions created for GOTeDNA:</u>

BeBOP_ID – Identifier for the protocols used obtained from (or inputted into) Better Biomolecular Ocean Practices (BeBOP)

GOTeDNAprotocol_ID - Common project ID to be used when lab and qPCR methods are identical, even for different species and locations.

GOTeDNAprotocolVersion - To be entered when project lab or field methods differ slightly. For example, when all field and lab methods are identical except for extraction kit, the protocol_ID would be the same, differing by version number. Rationale to combine data into a single protocol to be provided in protocolVersion_rationale (i.e., justifying validity of method by testing efficiency and/or peer-reviewed publication).

LClabel - When applicable (i.e., data collection on/by Indigenous peoples/lands), Local Contexts label for respecting and asserting Indigenous data sovereignty under the principles of OCAP (see: https://localcontexts.org/labels/biocultural-labels/). Under review.